

U.R NIKHIL BHAT

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ABOUT

Professional in High-Speed Design with 3.5+ years of experience in high-speed, mixed-signal board design, validation, and bring-up. Advancing expertise in [High-Speed Digital Engineering](#) at CU Boulder (Graduating Dec 2025) under Prof. Eric Bogatin. Preparing for impactful **Signal and Power Integrity (SI/PI)** roles starting Jan 2026, leveraging proven design experience with advanced skills in E.M modeling, simulation and measurement techniques.

WORK EXPERIENCE

Qualcomm, California

May 2025 – Aug 2025

Interim Hardware Intern

- Contributed to system-hardware design for next-gen Windows on Snapdragon, including backside PMIC placement beneath SiP shadow. Redesigned PDN with novel decoupling schemes (10× cost, 4× area gains) while meeting impedance mask.

CoreEL Technologies, Bangalore

Oct 2020 – Dec 2023

Senior Hardware Engineer

May 2023 – Dec 2023

- Re-engineered rugged Avionic Computers for a Fighter Jet with 4 cards and 2 power modules, achieving a 1000% boost in processing, memory, and communication speeds, a 20% power reduction, and 50% less backplane space usage.

Hardware Design Engineer

Oct 2020 – Apr 2023

- Developed a High-speed (12.5 Gbps JESD204B) Waveform/Timing board for an X-band Radar system.
- Engineered compact (15×7cm) VHF/UHF Handheld radio (1 km range, <4 dB NF, 1W Class D speaker, <-60 dBc emissions).

EDUCATION

University of Colorado Boulder, Boulder, USA

Aug 2024 – Present

- Master's Electrical Engineering (Subplan: [High-Speed Digital Engineering](#)) GPA – 3.9/4.0

Relevant Courses: [High-Speed Digital Design](#), [EM Signal Modelling:Ansys](#), [PCB Design & Mfg](#), [Computational EM for SI](#)

Nitte Meenakshi Institute of Technology, Bangalore, India

Aug 2016 – Aug 2020

Bachelor of Engineering (Electronics and Communication Engineering)

PROJECTS

[S-Parameter Analysis and Circuit Modelling:](#) Analyzed S-parameters in time and frequency domains, extracted Single, mixed-mode S- parameters, and developed an equivalent circuit model with bandwidth characterization up to 10 GHz.

[Post-Layout SI/PI Analysis:](#) Performed Comprehensive Post-layout SI/PI analysis(Skew/Length Matching, Crosstalk, Insertion Loss, Impedance Control, Timing - Eye Diagram, Jitter, Resonant and EMI) on Open Source Panther+ Micro Server Card.

[Impact of PCB Cavity Resonance on USB 3.0:](#) Validated return-via fencing to control plane-pair (GND–GND) modes that cause insertion-loss suck-outs by correlating lab data with post-layout EM/SI S-parameter extraction (Ansys HFSS); built breakout variants and a low-cost signal-generator board; journal submission underway.

[High-Speed Serial Links - Modeling & Equalization:](#) Used S-parameter-based sims (impulse/pulse response, PRBS, eye diagrams) and applied FFE to reopen eyes under ~20 dB insertion loss; quantified max throughput for stub/no-stub paths.

[Other SI/PI Projects:](#) Post and Pre-layout analysis: Crosstalk, TDR, High-speed links, Differential pair, PDN, DCR etc

[Sentinel-Fall: Privacy-Preserving Fall Detection System:](#) Developed a smart radar-based fall detection system with BLE connectivity winning a \$200 PCB fabrication coupon from Hackster.io.

SKILLS

SI/PI Skills – SI/PI Validation, Verification & Post-Layout Analysis. NRZ/PRBS/PAM4 Signaling. Time/Frequency-Domain Analysis. Transmission Line, High-Speed Connector & SerDes Channel Modeling & Extraction. PDN Design, Decoupling Strategies & Optimization. S-Parameter Extraction, Validation & Correlation. IBIS/IBIS-AMI & SPICE Simulations.

EDA & Simulation Tools – Ansys (HFSS 3D, 2D, SIwave, AEDT), HyperLynx, Keysight ADS, Matlab, Python. Cadence OrCAD, Altium Designer, Allegro Viewer, LTspice.

Hardware Design Skills – Analog/Digital & Mixed-Signal Circuit Board Design/Architecture.Signal Conditioning, Power (DC-DC & AC-DC), EMI/EMC Compliance, Schematic/Layout Review, Component Selection, Bring-Up, Testing & Debugging.

Controllers: SoC, SiP, SoM, Microprocessors, Microcontrollers, Xilinx FPGA

Test/Measurement Device – High B/W Oscilloscope, Signal Generator, TDR, VNA, Spectrum Analyzers, Logic Analyzer.

Interfaces/Protocols/Standards: CAN, I2C, I2S, LVDS, SerDes PCIe (Gen1 - Gen 5), Ethernet, HDMI, SATA, RS422, USB (1.0 - 3.0), SPI, UART, DDR3, LPDDR, DDR4, Wireless protocols and Bluetooth.